Change-Point Detection: MOSUM Methods

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Contents

1. Introduction
2. MOSUM statistic
3. Change-point estimators
4. Variance estimation
5. Choice of bandwidth

Introduction

In financial time series, structural breaks frequently occur due to factors such as policy changes, market shocks, shifts in investor sentiment, or unexpected events (e.g., financial crises, pandemics, or wars). These disruptions can lead to changes in statistical properties such as the mean, volatility, and the correlation structure of asset returns.

This implies that the data before and after a change-point no longer share the same model parameters. Continuing to apply a single model across such regimes may result in misestimation and increased risk exposure.

The main objectives are:

- Has a change occurred?
- If yes, where is the change?
- How certain are we about the change-point location?

Background

Change-point analysis can broadly be classified into two types:

- **Online (Sequential) Detection**: Monitoring for changes as the data is being observed.
- **Offline Detection**: Detecting changes after the full dataset has been collected.

The method to be introduced is an offline procedure designed to detect changes in the mean, known as MOSUM, short for Moving Sums.

The advantages of MOSUM:

- Easy interpretation and lend themselves naturally for visual inspection of structural changes.
- Can accommodate multiscale extensions of the MOSUM procedure for the improved adaptivity.
- Does not rely on any distributional assumption of the underlying data generating process.

MOSUM statistic

$$\underbrace{X_{k-G+1},\ X_{k-G+2},\ \ldots,\ X_k}_{\frac{1}{G}\sum_{t=k-G+1}^{k} X_t} \quad \longleftrightarrow \quad \underbrace{X_{k+1},\ X_{k+2},\ \ldots,\ X_{k+G}}_{\frac{1}{G}\sum_{t=k+1}^{k+G} X_t}$$

Consider observations $X_1, \ldots, X_n$ drawn independently from a distribution with the same mean as an example. By the Law of Large Numbers, it follows that

$$\frac{1}{G}\sum_{t=k+1}^{k+G} X_t - \frac{1}{G}\sum_{t=k-G+1}^{k} X_t \approx 0$$

for a sufficiently large **summation bandwidth** $G > 0$. If, on the other hand, there is a change in the mean of height d at time point k, then

$$\frac{1}{G}\sum_{t=k+1}^{k+G} X_t - \frac{1}{G}\sum_{t=k-G+1}^{k} X_t \approx d.$$

MOSUM statistic

Based on this observation, the following **MOSUM statistic** provides a good tool for change-point detection:

$$T_G := \max_{1 \le k \le n} \frac{|T_G(k)|}{\hat{\sigma}_k} \tag{1}$$

with the **MOSUM detector**

$$T_G(k) = T_G(k; X) := \sqrt{\tfrac{G}{2}} \left(\tfrac{1}{G} \sum_{t=k+1}^{k+G} X_t - \tfrac{1}{G} \sum_{t=k-G+1}^{k} X_t \right), \quad k = G, \ldots, n-G \tag{2}$$

and a local estimator $\hat{\sigma}_k^2$ of the innovation variance.
The values $T_G(k)$ for k at the left and right boundaries can be calculated adopting a CUSUM-type **boundary extension**:

$$T_G(k) := \sqrt{\frac{2G}{k(2G-k)}} \sum_{t=1}^{k} \left(\bar{X}_{(1,2G)} - X_t \right), \quad k = 1, \ldots, G-1, \tag{3}$$

with $\bar{X}_{(1,2G)} := (2G)^{-1} \sum_{t=1}^{2G} X_t$ and similarly for $k = n-G+1, \ldots, n$.

MOSUM statistic

The statistical model underlying the MOSUM procedure is a classical change-point location model (Eichinger and Kirch, 2018).

$$X_t = f_t + e_t = \sum_{j=1}^{N+1} \mu_j \mathbb{I}\{k_{j-1} < t \leq k_j\} + e_t, \quad t = 1, \ldots, n. \tag{4}$$

The piecewise constant deterministic signal f_t has N change-points at k_j, $j = 1, \ldots, N$ (with the notational convention $k_0 := 0$ and $k_{N+1} := n$), and the centred model innovations e_t are assumed to be independent and identically distributed (i.i.d). Note that the MOSUM detector from (2) is decomposed as

$$T_G(k; X) = T_G(k; f) + T_G(k; e). \tag{5}$$

We call $T_G(k; f)$ the **MOSUM signal** and $T_G(k; e)$ the noise term.

MOSUM statistics

To make this less abstract, consider the Figure 1.

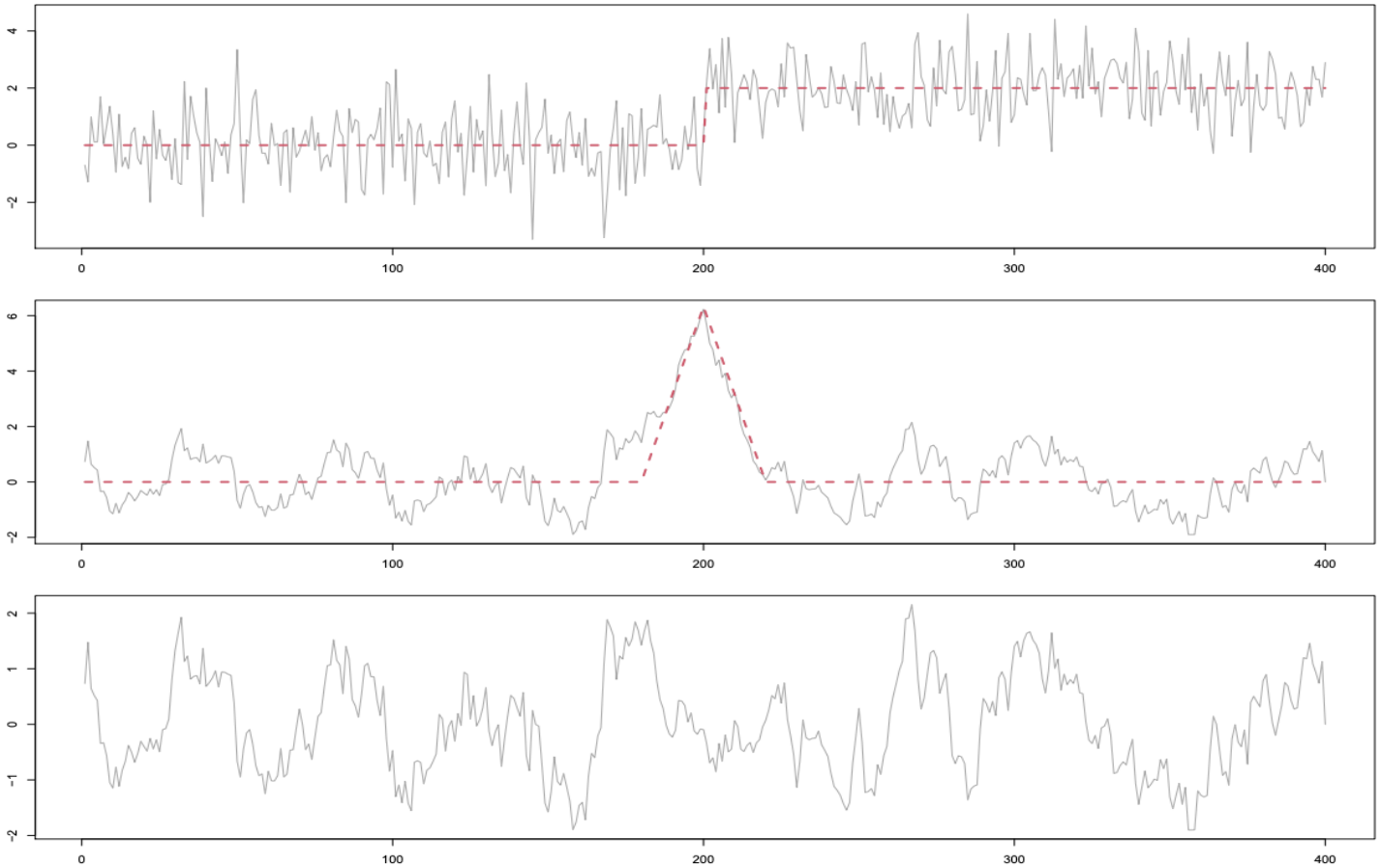

Figure 1. Top: A time series $X_1, \ldots, X_n$ of length $n = 400$ with one mean change of height $d = 2$ at time $k = 200$. The corresponding step signal f_t is overlaid as dashed line. Middle: The MOSUM detector $T_G(k; X)$ for $k = 1, \ldots, n$, where the MOSUM signal $T_G(k; f)$ is overlaid as dashed line. Bottom: $T_G(k; e) = T_G(k; X) - T_G(k; f)$.

MOSUM statistics

To derive a MOSUM-based test for detecting changes in the mean, one can utilize the following asymptotic result (The derivation is in Section C of the Appendix in Meier et al. (2021)).

For bandwidths $G = G(n)$ satisfying $G \to \infty$ as $n \to \infty$ while $G/n \to 0$, we have:

$$a_G T_G - b_G \Rightarrow \Gamma_2 \quad \text{under } H_0 : \text{No mean change,}$$

where $\Rightarrow$ denotes the convergence in distribution, and Γ_2 is a Gumbel-distributed random variable with $P(\Gamma_2 \leq z) = \exp\left(-2\exp(-z)\right)$. a_G and b_G are sequences of properly chosen scaling and shifting factors depending only on the sample size n and the bandwidth G.

MOSUM statistics

This result motivates a test at asymptotic level α, which rejects the null hypothesis

$$H_0 : N = 0 \quad \text{against} \quad H_1 : N > 0$$

when the MOSUM statistic T_G exceeds the asymptotic critical value:

$$C_{n,G}(\alpha) := \frac{b_G + Q_{1-\alpha}(\Gamma_2)}{a_G}, \tag{6}$$

where $Q_{1-\alpha}(\Gamma_2)$ denotes the $(1-\alpha)$-quantile of the Gumbel distribution. The corresponding p-value associated with the test statistic t is given by:

$$p_{n,G}(t) = 1 - \exp\left(-2\exp(b_G - a_G t)\right).$$

MOSUM statistics

The MOSUM statistic detects changes in the mean of a time series by comparing adjacent windows. It is based on a piecewise constant model with change-points at unknown locations. Under the null hypothesis of no change, the test statistic follows an asymptotic Gumbel distribution. A change is detected if the MOSUM statistic exceeds a critical value, or equivalently, if the p-value falls below the significance level.

How to identify the number and locations of the change points?

Change-point estimators

Reconsider Figure 1: a jump of the underlying step signal f_t results in a peak in the MOSUM signal, with the location of the jump coinciding with that of the local maximum of $|T_G(k; f)|$.

In practice, $T_G(k; f)$ is not observable and we have to work with the noisy MOSUM detector $T_G(k) = T_G(k; X)$ instead.

Therefore, it is natural to apply a threshold to the (scaled) absolute MOSUM detector and construct change-point estimators based on the local maximum of neighborhoods exceeding the threshold.

Change-point estimators

To elaborate, we consider *significant neighborhoods* (l, r) with $l \leq k \leq r$, such that

$$\frac{|T_G(k)|}{\hat{\sigma}_k} \geq C_{n,G}(\alpha) \quad \text{for } k = l, \ldots, r, \quad \text{and}$$

$$\frac{|T_G(k)|}{\hat{\sigma}_k} < C_{n,G}(\alpha) \quad \text{for } k = l-1, r+1.$$

Choose the maximal point within every significant environment

$$k^{\circ}_{(l,r)} = \arg\max_{l \leq k \leq r} \frac{|T_G(k)|}{\hat{\sigma}_k} \tag{7}$$

as a change-point estimator.

Change-point estimators

The MOSUM signal $T_G(k; f)$ behaves as a linear function on either side of a true change-point and will naturally cross the threshold. However, due to random noise, the detector may briefly drop below and then exceed the threshold again, leading to spurious significant neighborhoods and false change-point detections. This phenomenon, illustrated around time point 90 in Figure 2, highlights the risk of over-estimation.

Change-point estimators

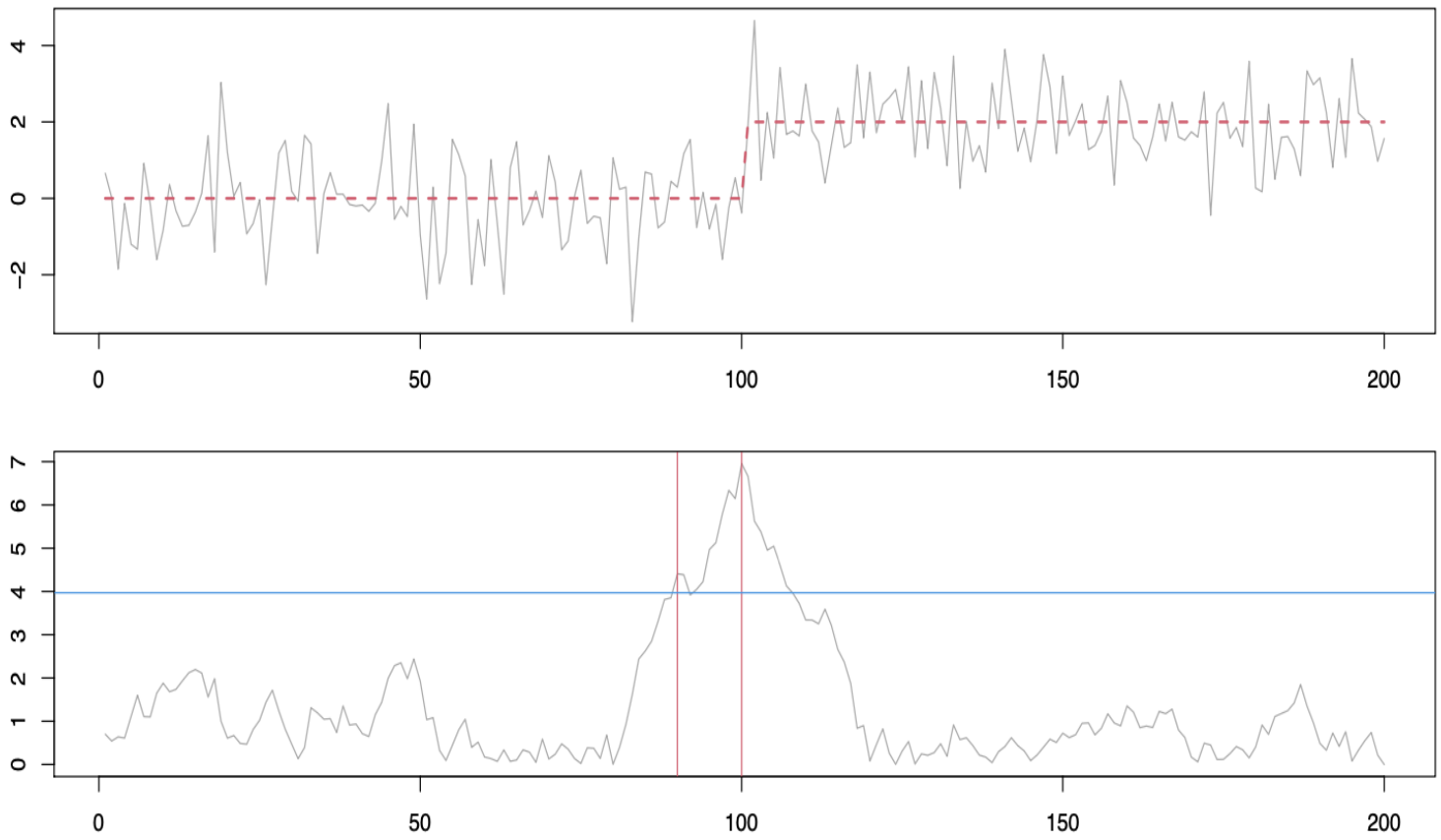

Figure 2. Top: A time series with one mean change at the location 100, where the underlying step signal f_t is overlaid as dashed line. Bottom: The corresponding absolute MOSUM detector $|T_G(k)|$ for $k = 1, \ldots, n$ with bandwidth $G = 20$. The critical value for level $\alpha = 0.05$ is denoted by a horizontal line, and maximal points within significant environments of $T_G(k)$ are highlighted by the vertical lines.

Change-point estimators

To mitigate this, the `mosum` R package implements two mathematical criteria to avoid such false positives: the ε-criterion and the η-criterion.

The ε-Criterion

The ε-criterion accepts a local maximum $k^{\circ}_{(l,r)}$ as a change-point only if the significant interval is sufficiently wide, i.e.,

$$r - l \geq \varepsilon G,$$

where $0 < \varepsilon < 1/2$ is fixed. This ensures that the signal is not merely a transient peak influenced by nearby changes, but a stable change-point candidate. Based on extensive simulation studies, $\varepsilon = 0.2$ is recommended as a default choice, which works well in many situations. This criterion becomes especially useful when the bandwidth G is sufficiently large, i.e., $G \geq 20$.

Change-point estimators

The η-Criterion

Let $\eta \geq 0$ be fixed. The η-criterion accepts a candidate k° as a change-point if it is the maximal point within its own neighborhood $[k^\circ - \eta G,\, k^\circ + \eta G]$, i.e.,

$$k^\circ = \arg \max_{k^\circ - \eta G \leq k \leq k^\circ + \eta G} \frac{|T_G(k)|}{\hat{\sigma}_k}.$$

Simulation studies suggest that $\eta = 0.4$ is a reasonable default, particularly when there is no prior knowledge about the mutual distance between change-points. As it is less conservative than the ε-criterion, the η-criterion is especially useful when the bandwidth is small.

Change-point estimators

Comparison of the ε- and η-Criterion.

- The ε-Criterion is relatively conservative, as it requires a whole neighborhood of length at least εG to exceed the threshold. This can be too restrictive in practice. For example, when the bandwidth is small (e.g., $G = 8$), even short significant environments of length 1 or 2 may indicate real changes and should not be discarded. See Figure 6 for an example of such a situation.
- Moreover, for larger bandwidths and certain jump signals, the detector may stay entirely above the threshold. In such cases, the ε-criterion only retains the global maximum, potentially missing multiple nearby change-points. See Figure 4 for an example of such a situation. By contrast, a well-chosen η-criterion can identify multiple peaks in the detector and thus reveal multiple changes more accurately.

Change-point estimators

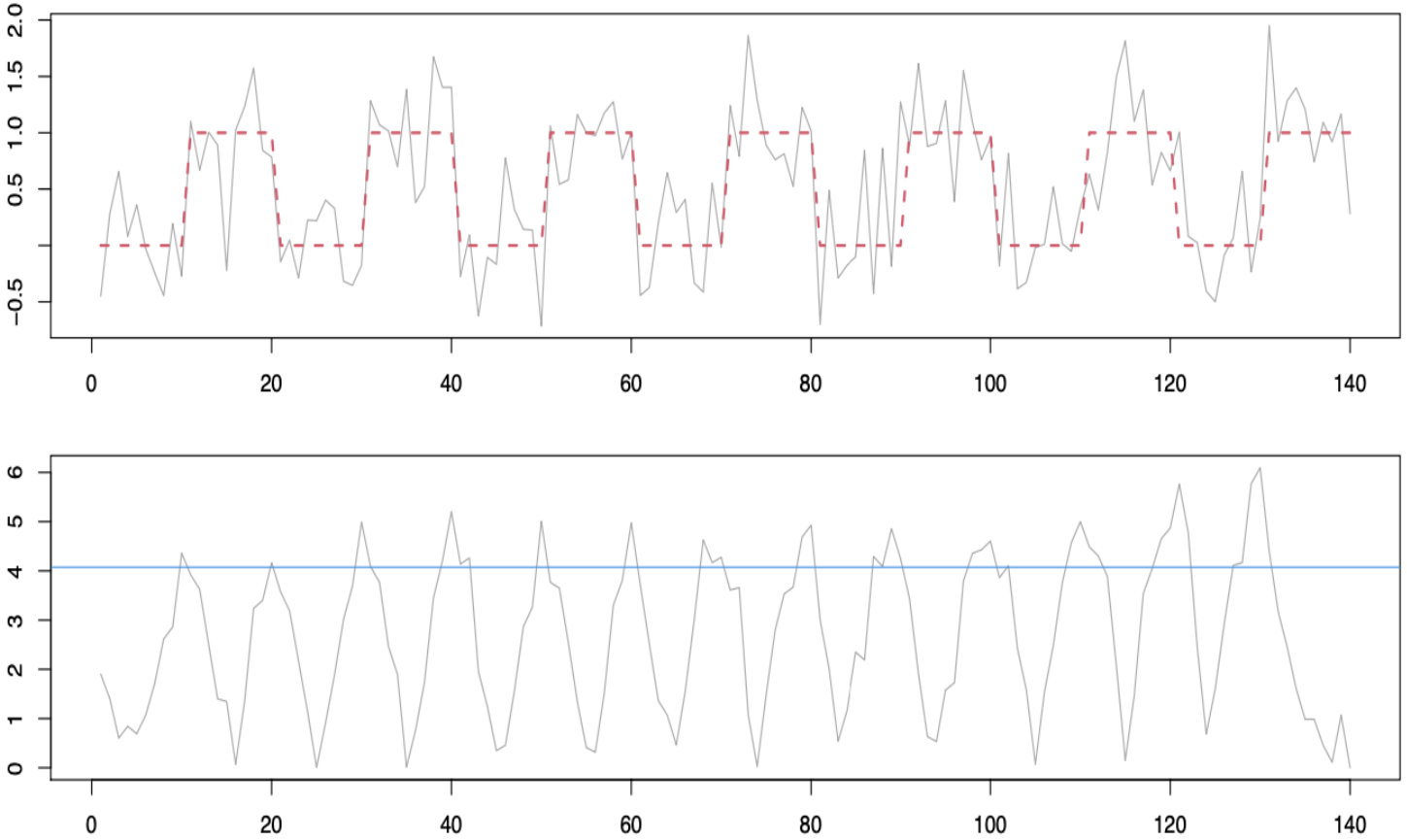

Figure 3. Top: A time series of length 140 with mean changes at locations 10, 20, . . . , 130, where the underlying step signal f_t is overlaid as dashed line. Bottom: The corresponding scaled absolute MOSUM detector $\sigma^{-1}|T_G(k)|$ with $G = 8$; the critical value for level $\alpha = 0.05$ is denoted by a horizontal line. The series is a realization of the model `teeth10` from Fryzlewicz (2014).

Change-point estimators

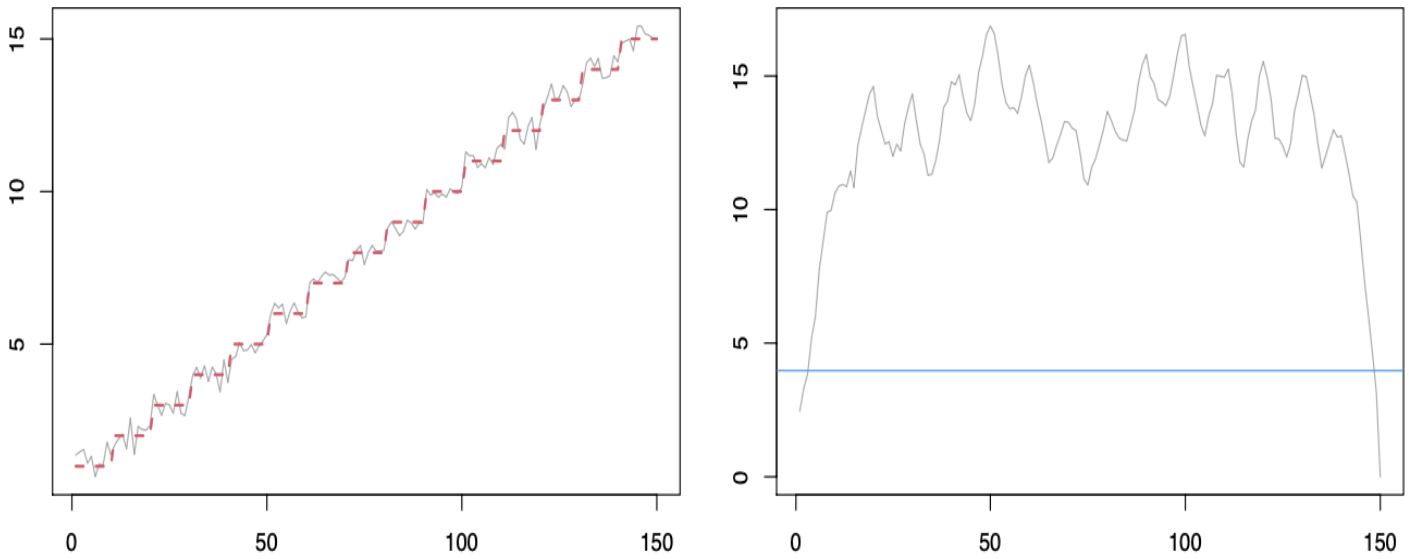

Figure 4. Left: A time series of length 150 with mean changes at locations 10, 20, . . . , 140, where the underlying step signal f_t is overlaid as dashed line. Right: The corresponding scaled absolute MOSUM detector $\sigma^{-1}|T_G(k)|$ with $G = 15$; the critical value for level $\alpha = 0.05$ is denoted by a horizontal line. The series is a realization of the model `stairs10` from Fryzlewicz (2014).

Variance estimation

Standard variance estimators such as

$$\frac{1}{n-1}\sum_{t=1}^{n}(X_t - \bar{X}_n)^2$$

are consistent in the absence of mean changes, but tend to systematically over-estimate variance when mean changes occur. This leads to poor performance in change-point detection.

A more robust approach is to use local variance estimators, such as the MOSUM-based estimator:

$$\hat{\sigma}^2_{(l,r)} := \frac{1}{r-l+1}\sum_{t=l}^{r}(X_t - \bar{X}_{(l,r)})^2,$$

where $\bar{X}_{(l,r)}$ is the sample mean of obsesrvations $X_l, X_{l+1}, ..., X_r$.

Variance estimation

Possible choices of the local variance estimator $\hat{\sigma}_k^2$ include:

$$\hat{\sigma}_k^2 = \frac{1}{2}\left(\hat{\sigma}^2_{(k-G+1,k)} + \hat{\sigma}^2_{(k+1,k+G)}\right) \tag{8}$$

$$\hat{\sigma}_k^2 = \min\left\{\hat{\sigma}^2_{(k-G+1,k)},\ \hat{\sigma}^2_{(k+1,k+G)}\right\} \tag{9}$$

$$\hat{\sigma}_k^2 = \max\left\{\hat{\sigma}^2_{(k-G+1,k)},\ \hat{\sigma}^2_{(k+1,k+G)}\right\} \tag{10}$$

The variance estimator in (9) is preferable for the detection of change-points at which not only the mean but also the variance undergoes changes, whereas the estimator in (10) is more robust in the sense that it will prevent the detection of spurious estimators in the presence of time-varying variance.

Variance estimation

When the error terms e_t exhibit weak dependence, the asymptotic validity of the test remains intact if the variance estimators $\hat{\sigma}_k^2$ are replaced with appropriate long-run variance estimators.

Estimator	Proposed by	Applicable When
Flat-top kernel estimator	Politis and Romano (1995)	Requires large bandwidth (e.g., $G \geq 50$)
Difference-based estimators	Tecuapetla-Gómez and Munk (2017); Dette et al. (2020); Axt (2019)	Suitable under weak dependence as an alternative to kernel estimators
Local MOSUM-based estimator	Meier et al. (2021)	Recommended for frequent change-points and small/medium samples
Threshold adjustment $C_{n,G}(\alpha) \cdot \log(n/G)^{\delta}$	Meier et al. (2021)	Used with local estimators; e.g., $\delta = 0.1$, to compensate for ignoring dependence

However, the first two estimators are not recommended in the presence of frequent change-points, as accurate estimation of the long-run variance is typically very difficult, and estimators based on small or medium-sized samples are often unreliable.

Choice of bandwidth

Smaller bandwidths can detect large jumps even near other change-points, while larger bandwidths may blur their locations due to signal contamination. With large bandwidths, the signal may flatten at the top, leading to arbitrary maxima across an interval, which hinders precise change-point localization. See Figure 5 (second and third panels).

On the other hand, large bandwidths can detect small isolated changes that small bandwidths may miss, especially in long stationary stretches (Figure 5, e.g., $k_1 = 100$ requires $G \geq 70$).

Similarly, symmetric bandwidths in the MOSUM detector may lack flexibility—e.g., when a small change is near a large one, small G may miss it, while large G suffers from contamination (Figure 6, top panel). This motivates the use of **asymmetric bandwidths** $\mathbf{G} = (G_l, G_r)$.

Choice of bandwidth

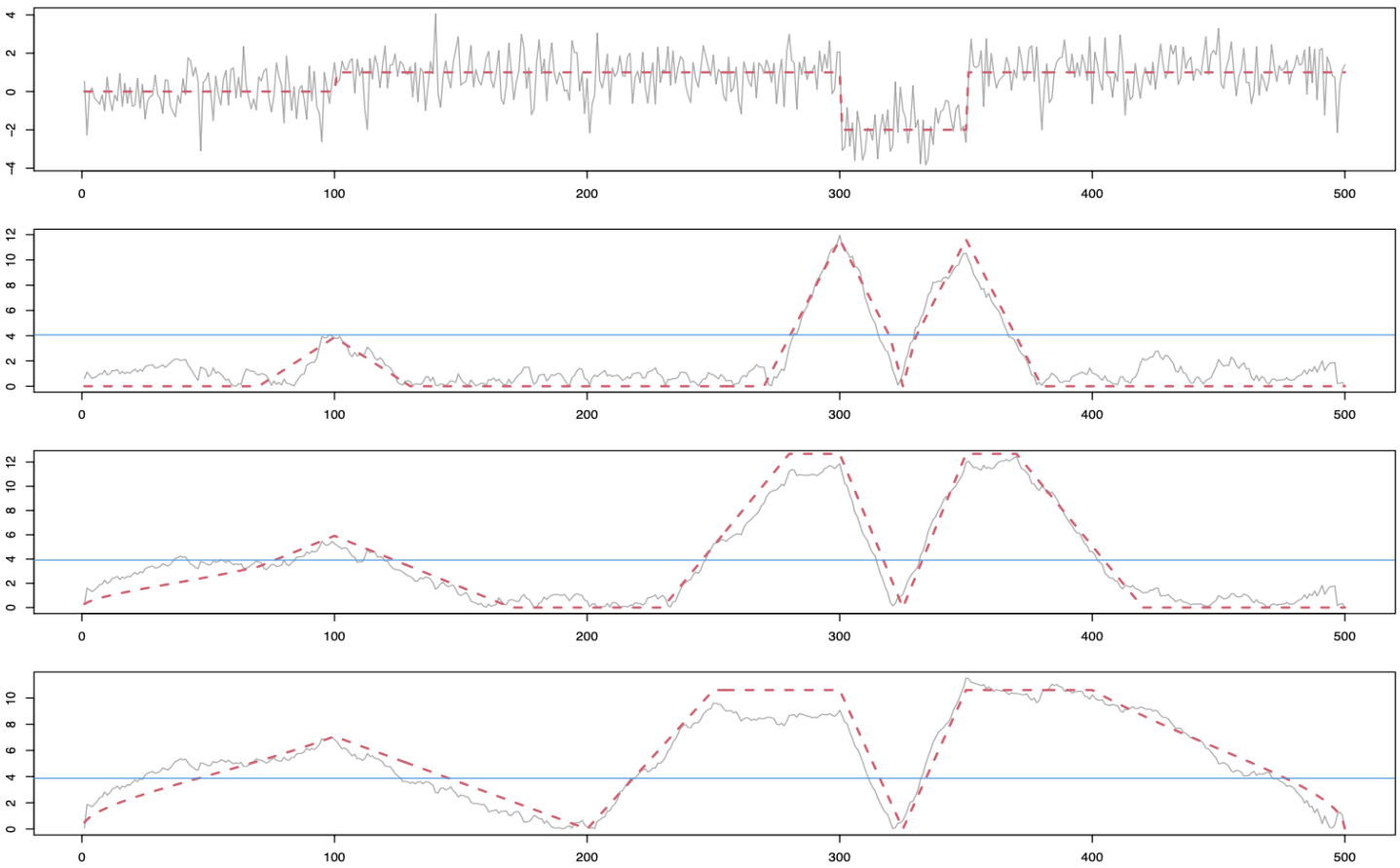

Figure 5. From top to bottom: A time series with mean changes at locations $k_1 = 100$, $k_2 = 300$ and $k_3 = 350$, where the underlying step signal f_t is overlaid as dashed line, and the absolute MOSUM detector $|T_G(k)|$ for $G = 30, 70$ and 100, where the absolute MOSUM signal $|T_G(k; f)|$ is overlaid as dashed line and the critical value at level $\alpha = 0.05$ is visualized by a solid horizontal line.

Choice of bandwidth

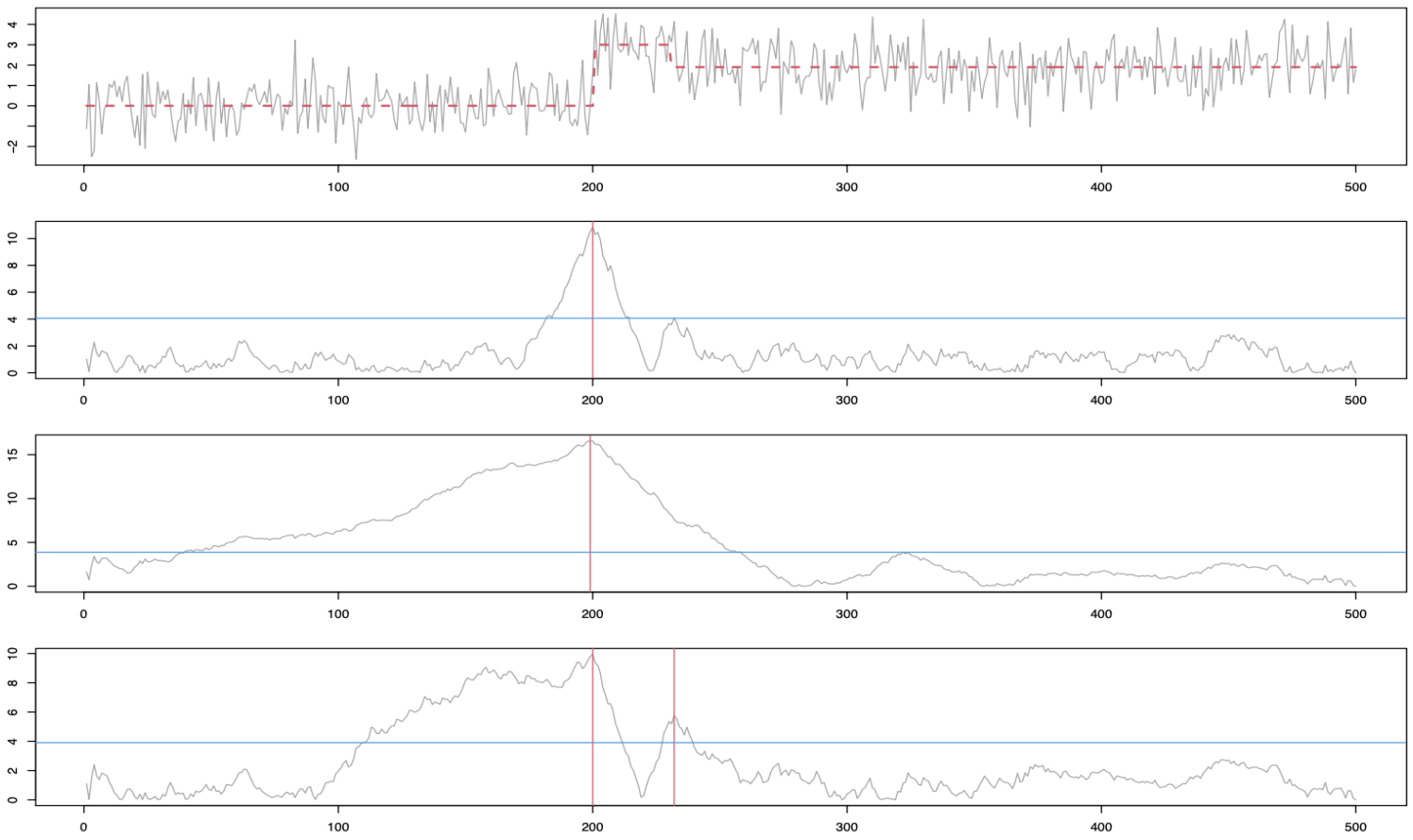

Figure 6. From top to bottom: A time series of length $n = 500$ with two mean changes at the locations 200 and 230 of respective height 3 and -1.1, and the corresponding absolute MOSUM detector $|T_G(k)|$ for $k = 1, \ldots, n$ with (symmetric) bandwidths $G = 30, 120$ and $\mathbf{G} = (30, 120)$. Critical value at level $\alpha = 0.05$ is visualized by a solid horizontal line and the change-points estimated according to the η-rule by solid vertical lines.

Asymmetric bandwidths

Let $\mathbf{G} = (G_l, G_r)$ be a bandwidth, possibly asymmetric (i.e., $G_l \neq G_r$). The **asymmetric MOSUM detector** is defined as

$$T_{\mathbf{G}}(k) = \sqrt{\frac{G_l \cdot G_r}{G_l + G_r}} \left(\frac{1}{G_r} \sum_{t=k+1}^{k+G_r} X_t - \frac{1}{G_l} \sum_{t=k-G_l+1}^{k} X_t \right), \; k = G_l, \ldots, n - G_r. \tag{11}$$

At the **left and right boundaries**, the values of $T_{\mathbf{G}}(k)$ can be defined by a CUSUM extension similar to (3):

$$T_{\mathbf{G}}(k) := \sqrt{\frac{G_l + G_r}{k(G_l + G_r - k)}} \sum_{t=1}^{k} \left(\bar{X}_{(1, G_l + G_r)} - X_t \right), \quad k = 1, \ldots, G_l - 1, \tag{12}$$

and analogously for $k = n - G_r + 1, \ldots, n$.

Asymmetric bandwidths

The critical value $C_{n,\mathbf{G}}(\alpha)$ for the asymmetric MOSUM test, as well as the corresponding *p-values* $p_{n,\mathbf{G}}$, can be obtained in a manner similar to the symmetric case; see Section C of the Appendix in Meier et al. (2021).
The extension of the change-point estimators to the asymmetric case is straightforward. The ε-criterion is adjusted to

$$r - l \geq \frac{\varepsilon}{2}(G_l + G_r), \tag{13}$$

and the η-criterion is adjusted to

$$k^\circ = \arg \max_{k^\circ - \eta G_l \leq k \leq k^\circ + \eta G_r} \frac{|T_{\mathbf{G}}(k)|}{\widehat{\sigma}_k}. \tag{14}$$

The MOSUM-based variance estimators can be defined with asymmetric bandwidths in a similar fashion as the MOSUM detector. When using asymmetric bandwidths, highly unbalanced pairs should be avoided due to their negative impact on both asymptotic properties and finite-sample performance. A common remedy is to bound $\max(G_l, G_r)/\min(G_l, G_r)$ from above.

Asymmetric bandwidths

In practice, a single bandwidth may lack flexibility to capture diverse changes, motivating the use of multiple bandwidths. However, this can lead to spurious or redundant estimates (multiple estimates relate to the same underlying true change point), thus necessitating a **merging/pruning** step . We introduce two procedures to refine candidate change-points obtained from the MOSUM method with multiple bandwidths.

Asymmetric bandwidths

Multiscale MOSUM procedure with bottom-up merging

Bottom-up merging is well-suited for long time series with heterogeneous change-points across multiple scales. However, in cases with asymmetric bandwidths or highly uneven distributions of change-points, its conservative strategy may lead to missed detections. See Messer et al. (2014) for further details.

Multiscale MOSUM procedure with localized pruning

Localized pruning enables the use of asymmetric bandwidths, which can be selected based on an information criterion. In the R package implementation, the Schwarz Criterion is adopted for this purpose. For further methodological details, see Messer et al. (2014). A discussion of the theoretical properties of the multiscale MOSUM procedure with localized pruning—particularly its consistency in estimating both the number and locations of change points under general conditions—can be found in Cho and Kirch (2022).

Bootstrap confidence intervals

Consider a set $\hat{k}_1, \ldots, \hat{k}_{\hat{N}}$ of change-point estimates returned by the multiscale MOSUM procedures or by the single-bandwidth MOSUM. Conditional on consistent estimation of the multiple change-points, both in their total number (such that $\hat{N} = N$) and locations, we can construct confidence intervals for the locations of change-points via bootstrapping. Below we provide the simplest version of the bootstrap scheme for i.i.d. innovations.

Bootstrap confidence intervals

Denote the (possibly asymmetric) detection bandwidths associated with $\hat{k}_j$ by $\hat{\mathbf{G}}_j = (\hat{G}_{lj}, \hat{G}_{rj})$, and let $I_j := \{\hat{k}_{j-1} + 1, \ldots, \hat{k}_j\}$ for $j = 1, \ldots, \hat{N} + 1$, where the convention $\hat{k}_0 := 0$ and $\hat{k}_{\hat{N}+1} := n$ is employed. Note that I_j represents all time points between the $(j-1)$th and the jth estimated change-points, so that the underlying step signal is assumed to be (approximately) constant within this interval. A bootstrap replicate $\{X_t^* : t \in I_j\}$ of the observations $\{X_t : t \in I_j\}$ can be obtained by drawing a random sample of size $|I_j|$ from $\{X_t : t \in I_j\}$ (with replacement). Repeating this for all $j = 1, \ldots, \hat{N} + 1$ yields a bootstrap replicate $\{X_1^*, \ldots, X_n^*\}$ of the observed time series.
Then a bootstrap replicate k_j^* of $\hat{k}_j$ is obtained as

$$k_j^* = \underset{\hat{k}_j - \hat{G}_{lj} + 1 \leq k \leq \hat{k}_j + \hat{G}_{rj}}{\arg\max} \left| T^*_{\hat{\mathbf{G}}_j}(k) \right|,$$

where $T^*_{\hat{\mathbf{G}}_j}(k) = T_{\hat{\mathbf{G}}_j}(k; X_1^*, \ldots, X_n^*)$ is defined as in (11), with X_t replaced by its bootstrap replicates X_t^*. From this, confidence intervals for the change-point locations can readily be computed.

Bootstrap confidence intervals

To elaborate, assume that for each change-point estimate $\hat{k}_j$, we have B bootstrap replicates $k^*_{j,b}$, $1 \leq b \leq B$. Denote by $M_j(\alpha)$ the empirical $(1-\alpha/2)$-quantile of $\{|k^*_{j,1} - \hat{k}_j|, \ldots, |k^*_{j,B} - \hat{k}_j|\}$. Then a **pointwise** $100(1-\alpha)\%$-confidence interval $K_j^{\mathrm{Pw}}(\alpha)$ for k_j can be constructed as

$$K_j^{\mathrm{Pw}}(\alpha) := \left[\hat{k}_j - M_j(\alpha),\, \hat{k}_j + M_j(\alpha)\right]. \tag{20}$$

To construct **uniform** confidence intervals for $k_1, \ldots, k_N$, we take into account the multiple testing problem that arises when considering multiple estimates, by computing $M(\alpha)$ as

$$M(\alpha) = \min\left\{c : \frac{1}{B}\sum_{b=1}^{B} \mathbb{I}\left(\max_{1\leq j\leq N}\left(\frac{\hat{d}_j^2}{\hat{\sigma}_j^2}|k^*_{j,b} - \hat{k}_j|\right) \leq c\right) \geq 1-\alpha\right\}, \tag{15}$$

where $\hat{d}_j = \bar{X}_{(\hat{k}_{j+1},\hat{k}_{j+1})} - \bar{X}_{(\hat{k}_{j-1}+1,\hat{k}_j)}$ denotes the estimated height of the jump at $t = \hat{k}_j$, and

Bootstrap confidence intervals

$$\hat{\sigma}_j^2 = \frac{1}{\hat{k}_{j+1} - \hat{k}_{j-1} - 2} \left(\sum_{t=\hat{k}_{j+1}}^{\hat{k}_{j+1}} (X_t - \bar{X}_{(\hat{k}_{j+1}, \hat{k}_{j+1})})^2 + \sum_{t=\hat{k}_{j-1}+1}^{\hat{k}_j} (X_t - \bar{X}_{(\hat{k}_{j-1}+1, \hat{k}_j)})^2 \right).$$

the pooled innovation sample variance. Then, uniform $100(1-\alpha)\%$-confidence intervals for $k_1, \ldots, k_N$ are given by

$$K_j^{\mathrm{unif}}(\alpha) := \left[\hat{k}_j - \frac{M(\alpha)\hat{\sigma}_j^2}{\hat{d}_j^2}, \hat{k}_j + \frac{M(\alpha)\hat{\sigma}_j^2}{\hat{d}_j^2} \right], \quad j = 1, \ldots, N. \tag{17}$$

Note that for each $j = 1, \ldots, N$, if either of the two endpoints of $K_j^{\mathrm{Pw}}(\alpha)$ falls outside the detection interval $(\hat{k}_j - \hat{G}_{lj}, \hat{k}_j + \hat{G}_{rj}]$, we trim off the confidence intervals so that $K_j^{\mathrm{Pw}}(\alpha) \subset (\hat{k}_j - \hat{G}_{lj}, \hat{k}_j + \hat{G}_{rj}]$; a similar step is taken for the uniform confidence intervals $K_j^{\mathrm{unif}}(\alpha)$.

Application

Salinas et al. (2020) propsed a probabilistic forecasting method based on autoregressive recurrent neural networks, capable of learning a global model across thousands of related time series and generating accurate and calibrated uncertainty forecasts.

Building on DeepAR, DeepCAR (Jungbluth and Lederer, 2023) introduces a MOSUM-based change point detection mechanism, which avoids selecting batches that contain abrupt structural changes during training. This approach significantly enhances the model's ability to adapt to structural variations in the data.

Compared with both DeepAR and the baseline naïve model, DeepCAR demonstrates a clear advantage in root mean square error (RMSE) when forecasting the Federal Reserve Board One Year Treasury Rate time series.

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